

Seventh Semester B. Tech. (Electronics and Communication Engineering) Examination

DIGITAL IMAGE AND VIDEO PROCESSING

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions are compulsory and carry marks as indicated.
- (2) Assume suitable data wherever necessary.
- (3) Illustrate your answer with neat sketches.
- (4) Use of Non – programmable calculator is permitted.

1. (a) Consider an image segment shown below in Fig. Q. 1(a) and compute length of the shortest–4, shortest–8 and shortest–m paths between pixels p and q where, $V = \{80, 83, 102\}$. If a certain path doesn't exist explain its reason.

83(a)	102	80	102	80
80	83	80	84	102(b)
102	102	83	91	84
83	80	84	102	91
91	102	80	83	84

Fig. Q. 1 (a)

Find the distance D_8 and D_e between pixel p and q. 6(CO1)

- (b) An object is 15 cm wide and 12 cm in height is imaged with a sensor of size $8.8 \times 6.6 \text{ mm}^2$ from a distance of 10 m. Calculate the required optimum focal length. 4(CO1)
2. (a) A transformation kernel is given below ; Conclude whether this is orthogonal transform.

$$A = \frac{1}{3} \begin{bmatrix} -1 & 2 & -2 \\ -2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} \quad 3(\text{CO3})$$

- (b) Apply DCT to the image given in Fig. 2 (b) and comment on its perfect reconstruction by taking IDCT.

2	0	0	0
1	2	1	1
1	0	0	1
1	1	2	1

Fig. 2 (b)

7(CO3)

3. (a) Realize the Laplacian operation given by equation to an image in Fig. 3 (a). Comment on the result image.

$$\nabla^2 f(x, y) = f(x+1, y) + f(x-1, y) + f(x, y+1) + f(x, y-1) - 4f(x, y)$$

12	4	4	4
8	12	0	0
6	12	6	6
12	8	8	6

Fig. 3 (a)

6(CO3)

- (b) Discuss the steps for applying ideal highpass filter in frequency domain to a grayscale image.

4(CO3)

4. (a) Compute threshold value using Otsu's method for an image of size 4×4 having normalized histogram shown in Fig. 4 (a).

<i>i</i>	0	1	2	3	4	5	6	7
<i>n_i</i>	1	2	2	3	2	3	2	1
<i>p_i</i>	0.1	0.1	0.1	0.2	0.1	0.2	0.1	0.1

6(CO3)

- (b) Discuss the frames to represent Group of Pictures (GOP).

4(CO2)

5. (a) Illustrate with neat sketch and formulate how angular frequency depends on viewing distance.

5(CO2)

- (b) Describe the fundamental parameters for optical flow computation.

5(CO4)

6. (a) Explain pixel based method of motion estimation along with its limitations.
5(CO4)
- (b) Develop an image processing system which will accept an input gray scale image shown in Fig. 6 b-1 and result in output image shown in Fig. 6 b-2. Draw the block diagram of the system and explain its functioning you have proposed.

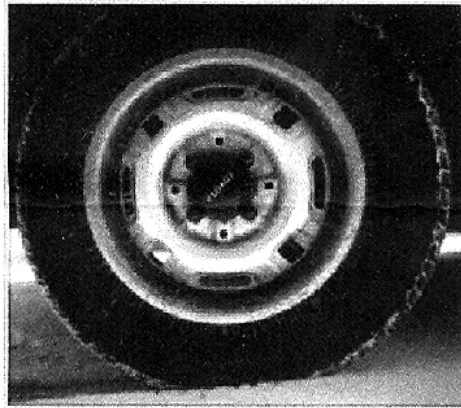


Fig. 6 b-1



Fig. 6 b-2

5(CO5)

